ENDPOINT TECHNOLOGIES PLC

Secure Network Design & Implementation Report

Project Title	Secure Network Design & Implementation — CCNA Level with Security Controls
Organisation	Endpoint Technologies PLC, Dhaka, Bangladesh
Prepared By	Asfi Ahmed — Cybersecurity Analyst (Fresher)
Tool Used	Cisco Packet Tracer 8.2.1
Frameworks	ISO/IEC 27001:2022 | NIST Cybersecurity Framework
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1. Executive Summary

This report documents the design, implementation, and verification of a secure enterprise network for Endpoint Technologies PLC — a mid-sized software development and IT services organisation based in Dhaka, Bangladesh. The network was simulated using Cisco Packet Tracer 8.2.1 and implements CCNA-level configurations with enterprise-grade security controls.

The implementation directly addresses critical risks identified in the organisation's ISO/IEC 27001 Risk Assessment (Project 1), translating identified vulnerabilities into technical security controls. Every configuration decision is mapped to a specific risk, ISO 27001 Annex A control, and NIST Cybersecurity Framework subcategory.

Key achievements of this implementation include:

- Five isolated network segments (VLANs) eliminating lateral movement risk
- Router-on-a-Stick inter-VLAN routing with enforced access control policies
- DMZ architecture isolating public-facing servers from internal infrastructure
- Extended ACLs enforcing granular department-level access control
- SSH-only remote management with RSA 2048-bit encryption
- DHCP automation with IP conflict prevention
- Port security with MAC address locking on all access ports
- Full device hardening including password encryption, banner warnings and unused port shutdown

2. Organisation Overview

Endpoint Technologies PLC is a mid-sized technology organisation operating in Dhaka, Bangladesh. The company provides web and mobile application development, API-based enterprise solutions, and internal SaaS platforms to its clients.

Attribute	Details
Organisation	Endpoint Technologies PLC
Location	Dhaka, Bangladesh
Employees	50–100 employees
Business	Web/mobile development, API solutions, SaaS platforms
Security Context	Handles sensitive client data, source code, financial records, and employee credentials

3. Project Objective

The primary objective of this project is to design and simulate a secure enterprise network that directly mitigates the risks identified in Project 1 (ISO 27001 Risk Assessment). The three core objectives are:

- Network Segmentation: Implement VLAN-based segmentation to eliminate lateral movement between departments
- Access Control: Deploy extended ACLs to enforce a strict security policy governing inter-departmental communication
- Security Hardening: Apply industry best practices to all network devices to reduce the attack surface

This project serves as the technical implementation layer of Endpoint Technologies PLC's Information Security Management System (ISMS), demonstrating that identified risks have been actively mitigated through concrete technical controls.

4. Network Topology

4.1 Logical Design

The network is designed around a hub-and-spoke topology with a single Cisco 2911 router acting as the central routing and security enforcement point. Two Cisco 2960 access layer switches serve the internal departments, and a dedicated DMZ switch isolates public-facing servers.

VLAN	Department	Network	Gateway	Switch
VLAN 10	IT Department	192.168.10.0/24	192.168.10.1	ETP-SW-FLOOR 1
VLAN 20	HR Department	192.168.20.0/24	192.168.20.1	ETP-SW-FLOOR 1
VLAN 30	DEV Department	192.168.30.0/24	192.168.30.1	ETP-SW-FLOOR 2
VLAN 40	Guest Network	192.168.40.0/24	192.168.40.1	ETP-SW-FLOOR 2
VLAN 50	DMZ	192.168.50.0/24	192.168.50.1	ETP-SW-DMZ

4.2 Device Inventory

Asset ID	Device Name	Model	Role	IP/Network
A1	ETP-ROUTER-01	Cisco 2911	Core Router	All VLANs
A2	ETP-SW-FLOOR1	Cisco 2960	Access Switch	VLAN 10, 20
A3	ETP-SW-FLOOR2	Cisco 2960	Access Switch	VLAN 30, 40
—	ETP-SW-DMZ	Cisco 2960	DMZ Switch	VLAN 50
A5	WEB-SERVER	Server-PT	Apache Web Server	192.168.50.10
A6	DB-SERVER	Server-PT	MySQL Database	192.168.50.20
A8	IT-PC-01	PC-PT	IT Workstation	DHCP — .10.x
A8	HR-PC-01	PC-PT	HR Workstation	DHCP — .20.x
A8	DEV-PC-01	PC-PT	DEV Workstation	DHCP — .30.x
A8	GUEST-PC-01	PC-PT	Guest Device	DHCP — .40.x

5. Implementation

5.1 VLAN Segmentation

VLANs were configured on both access layer switches to logically isolate each department. This directly mitigates Risk R52 (lateral movement due to flat network) identified in the ISO 27001 Risk Assessment.

```
! ETP-SW-FLOOR1 VLAN Configuration
vlan 10
  name IT
vlan 20
  name HR
vlan 30
  name DEV
vlan 40
  name GUEST

! Port assignments
interface FastEthernet0/1
  switchport mode access
  switchport access vlan 10
interface FastEthernet0/2
  switchport mode access
  switchport access vlan 20
```

5.2 Trunking (802.1Q)

Trunk ports were configured on the uplink interfaces of all switches to carry tagged VLAN traffic to the router. The 802.1Q standard was used for VLAN tagging.

```
! Trunk configuration on all switches
interface FastEthernet0/24
  switchport mode trunk

! Verification
show interfaces trunk
! Expected: Port Fa0/24 | Mode: on | Encapsulation: 802.1q | Status: trunking
```

5.3 Router-on-a-Stick (Inter-VLAN Routing)

A single Cisco 2911 router was configured with logical subinterfaces — one per VLAN — to enable controlled inter-departmental communication. Each subinterface acts as the default gateway for its respective VLAN.

```
! ETP-ROUTER-01 Subinterface Configuration
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0

interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
  ip access-group HR_POLICY in

interface GigabitEthernet0/1.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
  ip access-group DEV_POLICY out

interface GigabitEthernet0/1.40
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0
  ip access-group GUEST_POLICY in

interface GigabitEthernet0/2.50
  encapsulation dot1Q 50
  ip address 192.168.50.1 255.255.255.0
```

5.4 DMZ Architecture

A dedicated DMZ (Demilitarized Zone) was implemented on a separate router interface (GigabitEthernet0/2) to host the organisation's public-facing Web Server (A5) and Database Server (A6). The DMZ is logically isolated from all internal VLANs, ensuring that a compromise of the web server cannot be used as a pivot point to access sensitive internal systems.

Note on Packet Tracer Limitation:

Full stateful DMZ isolation (blocking DMZ-initiated connections while allowing return traffic) requires a stateful firewall such as a Cisco ASA or Palo Alto Networks NGFW. Packet Tracer does not support stateful inspection. In a production environment, this control would be enforced by a next-generation firewall. This limitation has been documented as a known gap and noted in the project's risk register.

5.5 Access Control Lists (ACLs)

Extended named ACLs were deployed on the router to enforce the organisation's security policy. ACLs were applied at the interface level to control traffic flow between VLANs.

Security Policy Matrix

Source	Destination	Action	Reason
IT (VLAN 10)	Any	ALLOW	IT has full administrative access
HR (VLAN 20)	DEV (VLAN 30)	DENY	Sensitive code separation required
HR (VLAN 20)	IT / DMZ	ALLOW	HR requires access to systems and servers
Guest (VLAN 40)	All Internal	DENY	Full guest isolation — internet only
DMZ (VLAN 50)	Internal*	DENY*	Servers cannot initiate internal connections

* Production implementation requires stateful firewall — see Section 5.4

```
! HR_POLICY — Blocks HR from DEV, blocks DMZ source
ip access-list extended HR_POLICY
  deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
  deny ip 192.168.50.0 0.0.0.255 any
  permit ip any any

! GUEST_POLICY — Full guest isolation
ip access-list extended GUEST_POLICY
  deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
  deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
  deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
  deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
  deny ip 192.168.50.0 0.0.0.255 any
  permit ip any any

! DEV_POLICY — Blocks DMZ source from reaching DEV
ip access-list extended DEV_POLICY
  deny ip 192.168.50.0 0.0.0.255 any
  permit ip any any
```

5.6 DHCP Configuration

The router was configured as a DHCP server for all four internal VLANs. Static IP exclusions were applied to reserve addresses for infrastructure devices including gateways and switch management interfaces.

```
! DHCP Exclusions (reserved for static devices)
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10

! DHCP Pools
ip dhcp pool VLAN10-IT
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 8.8.8.8

ip dhcp pool VLAN20-HR
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 8.8.8.8

(Repeat for VLAN30-DEV and VLAN40-GUEST)
```

5.7 Port Security

Port security was configured on all active access ports to prevent MAC flooding attacks and unauthorised device connections. Sticky MAC learning was enabled to automatically lock the connected device's MAC address to its port.

```
! Port Security Configuration (applied to all active access ports)
interface FastEthernet0/1
  switchport mode access
  switchport port-security
  switchport port-security maximum 2
  switchport port-security violation restrict
  switchport port-security mac-address sticky

! Disable all unused ports
interface range FastEthernet0/3-23
  shutdown
```

5.8 Device Hardening

All network devices were hardened in accordance with industry best practices and ISO 27001 requirements. The following controls were applied to both switches:

```
! Enable secret (MD5 hashed)
enable secret ETP@Strong2026

! Encrypt all plaintext passwords
service password-encryption

! Console access control
line console 0
  password cisco123
  login

! Legal warning banner
banner motd #
Unauthorized access to Endpoint Technologies PLC network
is strictly prohibited. All activities are monitored and logged.
#
```

5.9 SSH Remote Management

Telnet was completely disabled on all managed devices. SSH version 2 with RSA 2048-bit encryption was configured to provide secure encrypted remote management access. Each switch was assigned a dedicated management VLAN interface.

```
! SSH Configuration (ETP-SW-FLOOR1)
hostname ETP-SW-FLOOR1
ip domain-name endpointtech.local
crypto key generate rsa                    ! Choose 2048 bits
ip ssh version 2
username admin privilege 15 secret Admin@ETP2026

line vty 0 4
  transport input ssh
  login local

! Management interface
interface Vlan10
  ip address 192.168.10.254 255.255.255.0
  no shutdown
ip default-gateway 192.168.10.1

! ETP-SW-FLOOR2 management on VLAN 30
```

```
interface Vlan30
  ip address 192.168.30.254 255.255.255.0

! ETP-SW-DMZ management on VLAN 50
interface Vlan50
  ip address 192.168.50.254 255.255.255.0
```

6. Test Results & Verification

All security controls were verified through systematic connectivity testing. The following test matrix documents the expected and actual results for every traffic flow in the network.

Source	Destination	Expected	Result	Control Applied
IT-PC-01	HR-PC-01	ALLOW	✅ PASS	No restriction on IT
IT-PC-01	DEV-PC-01	ALLOW	✅ PASS	No restriction on IT
IT-PC-01	WEB-SERVER	ALLOW	✅ PASS	IT has full DMZ access
IT-PC-01	DB-SERVER	ALLOW	✅ PASS	IT has full DMZ access
HR-PC-01	IT-PC-01	ALLOW	✅ PASS	HR → IT permitted
HR-PC-01	DEV-PC-01	DENY	✅ PASS	HR_POLICY deny rule
HR-PC-01	WEB-SERVER	ALLOW	✅ PASS	HR → DMZ permitted
GUEST-PC-01	IT-PC-01	DENY	✅ PASS	GUEST_POLICY deny rule
GUEST-PC-01	HR-PC-01	DENY	✅ PASS	GUEST_POLICY deny rule
GUEST-PC-01	DEV-PC-01	DENY	✅ PASS	GUEST_POLICY deny rule
GUEST-PC-01	WEB-SERVER	DENY	✅ PASS	GUEST_POLICY deny rule
SSH from IT-PC	ETP-SW-FLOOR 1	ALLOW	✅ PASS	SSH v2 RSA 2048-bit
DHCP Request	All VLANs	ALLOW	✅ PASS	4 DHCP pools configured

All 13 test cases passed. No unexpected traffic flows were observed during testing.

7. ISO/IEC 27001 Control Mapping

The following table maps each implemented security control to the corresponding ISO/IEC 27001:2022 Annex A control and the risk it mitigates from the Project 1 Risk Assessment.

Control Implemented	ISO 27001 Annex A	Risk Mitigated	Implementation
VLAN Segmentation	A.13.1.3 — Segregation in Networks	R52 — Lateral movement	5 VLANs isolating all departments
ACL Access Control	A.9.4 — Access Control	R43 — ACL misconfiguration	HR_POLICY, GUEST_POLICY, DEV_POLICY
DMZ Architecture	A.13.1.3 — Segregation	R13 — Web app vulnerabilities	Dedicated VLAN 50 for servers
SSH Encryption	A.10.1 — Cryptography	R01 — Weak credentials	RSA 2048-bit, SSHv2 only
Port Security	A.11.2 — Physical Security	R04 — MAC flooding	Sticky MAC, restrict violation mode
Device Hardening	A.9.4.2 — Secure Log-on	R01 — Weak credentials	Enable secret, password encryption
Banner MOTD	A.9.4.2 — Secure Log-on	R01 — Unauthorised access	Legal warning on all devices
Unused Port Shutdown	A.12.6 — Vulnerability Mgmt	R04 — Unauthorised access	fa0/3-23 shutdown on all switches
DHCP Automation	A.13.1.1 — Network Controls	IP conflict risk	4 DHCP pools with exclusions
Guest Isolation	A.13.1.3 — Segregation	R10 — Weak WiFi/Guest access	VLAN 40 bidirectionally isolated

8. NIST Cybersecurity Framework Mapping

Function	Category	Subcategory	Implementation
Identify	Asset Management	ID.AM-1	17 network assets inventoried and documented
Identify	Risk Assessment	ID.RA-1	50+ risks identified with likelihood/impact scoring
Protect	Access Control	PR.AC-4	VLAN segmentation + ACL least privilege enforcement

Protect	Access Control	PR.AC-3	SSH-only remote management on all devices
Protect	Data Security	PR.DS-2	RSA 2048-bit SSH encryption for data in transit
Protect	Protective Tech	PR.PT-3	Unused ports disabled — least functionality principle
Protect	Protective Tech	PR.PT-4	ACLs protecting communications and control networks
Detect	Monitoring	DE.CM-7	Port security monitors for unauthorised device connections
Respond	Mitigation	RS.MI-1	VLAN isolation contains potential breach to one segment
Recover	Recovery Plan	RC.RP-1	BCDR plan documented in Project 5

9. Known Limitations & Production Recommendations

The following limitations were identified during implementation. These are inherent constraints of the Packet Tracer simulation environment and would be addressed in a production deployment.

Limitation	Current State	Production Recommendation
Stateful DMZ Isolation	Router ACLs cannot differentiate initiated vs return traffic	Deploy Cisco ASA or Palo Alto NGFW for stateful inspection
Management VLAN	Management IPs on production VLANs (10, 30, 50)	Dedicated Management VLAN 99 with strict ACL access
SSH Access Control	SSH accessible from multiple VLANs	ACL restricting SSH access to IT VLAN only (port 22)
No Syslog	No centralised log collection configured	Deploy syslog server and configure all devices to forward logs
Single Router	Single point of failure at core router	HSRP with redundant router for high availability
No IDS/IPS	No intrusion detection configured	Deploy Cisco IDS/IPS or Snort for traffic analysis

10. Conclusion

This project successfully demonstrates the design and implementation of a secure enterprise network for Endpoint Technologies PLC. The implementation translates 10 critical risks identified in the ISO 27001 Risk Assessment into concrete, verifiable technical controls — proving the ability to bridge GRC methodology with hands-on network engineering.

The network achieves the following security outcomes:

- Complete departmental isolation through VLAN segmentation — eliminating lateral movement risk
- Enforced access control policy through extended ACLs — restricting inter-departmental communication to only what is necessary
- DMZ architecture protecting internal systems from public-facing server compromise
- Encrypted remote management through SSH version 2 with RSA 2048-bit keys
- Physical port security preventing unauthorised endpoint connections
- Fully hardened device configurations aligned with CIS benchmarks and ISO 27001 requirements

All 13 verification test cases passed. The network is fully operational and ready for portfolio presentation. Known limitations have been documented with production-grade remediation recommendations, demonstrating awareness of real-world enterprise security requirements beyond the simulation environment.

Prepared by: Asfi Ahmed | Cybersecurity Analyst (Fresher) | April 2026